Experiment 7: Ann's Rendezvous case

Aim: To recover email messages from packet capture and analyze the same.

Learning Outcomes:
After completion of this experiment, student should be able to analyze SMTP packets using wireshark.

Case:

After being released on bail, Ann Dercover disappears! Fortunately, investigators were carefully monitoring her network activity before she skipped town. "We believe Ann may have communicated with her secret lover, Mr. X, before she left," says the police chief. "The packet capture may contain clues to her whereabouts."
You are the forensic investigator. Analyze the given packet capture to answer following questions.

- Provide any online aliases or addresses and corresponding account credentials that may be used by the suspects under investigation.
- Who did Ann communicate with? Provide a list of email addresses and any other identifying information.
- Extract any transcripts of Ann's conversations and present them to investigators
- If Ann transferred or received any files of interest, recover them
- Are there any indications of Ann's physical whereabouts? If so, provide supporting evidence.

Network:
- Internal network:192.168.30.0/24
- DMZ: 10.30.30.0/24
- The Internet: 172.30.1.0/24 (for purpose of this case study)

Evidence:
Investigators provide you with a packet capture from Ann's home network. Evidence file is available on local server under ISAF\AnnCase.They also inform you that in the course of their monitoring, they have found that Ann's laptop has a MAC address of 00:21:70:4D;4F:AE.

Procedure
1. Open evidence file using wireshark.
2. Click statistics-→Protocol Hierarchy. Note the percentage of following protocols
   IP, UDP, TCP, DNS, bootstrap, SMTP, IMAP and HTTP.
3. Use wireshark display filter **eth.addr==00:21:70:4d:4f:ae and bootp.** Note MAC address, IP address, Host Name, lease time, DNS server address, Router Address, subnet mask, etc.
4. Use wireshark display filter **smtp.command_line.** Note number of EHLO messages, email addresses.

5. Use wireshark display filter **smtp.** Right click on first message and follow TCP stream. Is authentication detail encrypted?
6. Note username and password. Decode using base 64. Note the content of email.
7. Using networkminer software (available on local host), analyze given packet capture and note the following details related to email.
    a. To and from email ids
    b. Subject
    c. Message ids
    d. Timestamp values
    e. Attachment if any
8. Recover attachment and note the hash values of the same.
9. Upload your document on Student Portal by answering questions given at the end of case description.

Review questions:

1. Develop timeline for the given case.
2. Who did Ann communicate with? Provide a list of email addresses and any other identifying information
3. Are there any indications of Ann's physical whereabouts? If so, provide supporting evidence.